Quantification: 535_z20c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	16
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2821.75	2390.5	1756.86	px^2
major_axis_length_px	lacuna shape	84.376	84.2904	26.6285	px
minor_axis_length_px	lacuna shape	45.6176	40.768	18.0402	px
aspect_ratio	lacuna shape	1.98	1.8332	0.6428	unitless
eccentricity	lacuna shape	0.8076	0.836	0.1335	unitless
solidity	lacuna shape	0.8008	0.8089	0.1085	unitless
roots_count	headline	8.5625	9	3.1192	count
ring_length_r30_px	headline	302.938	285.5	83.7444	px
ring_length_r60_px	reported	628.062	651	195.864	px
owned_length_px	ownership-dependent	1025.84	1025.32	456.25	px
edge_count	ownership-dependent	46.375	42.5	20.893	count
mean_edge_length_px	ownership-dependent	22.3255	21.3237	4.5462	px
perimeter_px	normalised	278.054	252.664	111.253	px
ring_area_r30_px2	normalised	9153.12	9201	2338.97	px^2
ring_area_r60_px2	normalised	21102.8	21221.5	4616.18	px^2
in_frame_fraction_r30	normalised	0.9896	1	0.0415	unitless
in_frame_fraction_r60	normalised	0.9831	1	0.0664	unitless
ring_density_r30	normalised	0.0337175	0.0338105	0.00731804	px^-1
ring_density_r60	normalised	0.0297008	0.0289793	0.00635491	px^-1
roots_per_100px_perimeter	normalised	3.2592	3.2045	1.0713	per 100 px
ring_attached_length_r30_px	network v2	270.312	253	85.0496	px
ring_attached_length_r60_px	network v2	485.812	468.5	198.864	px
ring_length_w_r30_px	network v2	345.105	329.856	95.7299	px
ring_length_w_r60_px	network v2	714.994	744.254	223.273	px
sholl_crossings_r10	network v2	7.25	7	2.4358	count
sholl_crossings_r20	network v2	8.75	9	3.3166	count
sholl_crossings_r30	network v2	8.5625	9	3.4442	count
loop_count	loops, report only	1.9375	1	2.3796	count
loop_edge_share	loops, report only	0.124	0.1162	0.1243	share
ring_width_mean_r30_px	width	7.4424	7.539	0.9124	px

Per field
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measure	value	units
analysed_area_px2	1.00266e+06	px^2
total_skeleton_length_px	23612	px
canalicular_length_density_per_px	0.0235493	px^-1
junction_count	400	count
junction_density_per_px2	0.00039894	px^-2
lacuna_count	17	count

lacunae_per_px2	1.695e-05	px^-2
skeleton_component_count	380	count
mean_component_length_px	62.1368	px
median_component_length_px	21.5	px
field_length_density_w_per_px	0.0265547	px^-1
field_density_without_flagged_per_px	0.0230569	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	572	px
unattached_component_count	35	count
canalicular_length_density_attached_per_px	0.0235493	px^-1
field_length_density_with_unattached_per_px	0.0241198	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	31	count
loop_edge_share	0.0994	share
edge_count	1167	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6071	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.3137	px
width_pixels_used	20467	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	766	55.3886	19.9792	2.7723	0.9327	0.8815	1.5239	7.094	208.97	True
2	1417	59.2424	36.6108	1.6182	0.7862	0.7182	0.2448	34.1404	249.047	False
3	3380	83.9727	56.9872	1.4735	0.7345	0.8272	-0.9852	92.8941	716.001	False
4	532	31.4193	21.8844	1.4357	0.7175	0.9449	-0.5795	222.887	852.859	False
5	1982	85.686	32.1789	2.6628	0.9268	0.8251	-1.1473	243.598	120.229	False
6	1956	84.6082	30.6094	2.7641	0.9323	0.8815	-0.4607	416.865	210.624	False
7	7294	147.859	76.1316	1.9421	0.8573	0.7458	-0.0651	474.999	284.411	False
8	4856	115.922	57.0373	2.0324	0.8706	0.8161	-1.3108	461.194	574.536	False
9	3706	78.2414	69.9705	1.1182	0.4475	0.789	0.5114	581.897	502.58	False
10	2088	68.5717	40.0641	1.7115	0.8116	0.9198	0.595	570.565	144.556	False
11	2662	104.173	41.4719	2.5119	0.9173	0.6642	1.1534	613.184	873.72	False
12	2280	82.4791	35.9781	2.2925	0.8998	0.9205	-1.1889	632.433	317.15	False
13	1049	49.7107	28.8293	1.7243	0.8147	0.9468	1.1742	770.511	315.037	False
14	4401	92.4669	72.9809	1.267	0.6141	0.7373	1.4761	819.541	677.56	False
15	4198	98.8193	64.1556	1.5403	0.7606	0.7155	1.4566	847.201	603.76	False
16	2501	85.9153	42.1925	2.0363	0.8711	0.8016	1.0022	846.978	318.042	False
17	846	80.9285	22.7986	3.5497	0.9595	0.5599	-0.1838	864.301	863.892	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	0	51	210	44.2843	2	22.1421	
2	6	209	353	528.416	31	17.0457	
3	9	370	875	1400.05	49	28.5725	8.0097
4	5	257	610	658.541	31	21.2433	6.8989
5	11	454	988	2070.56	99	20.9147	7.9579
6	7	278	511	670.085	37	18.1104	8.7599
7	16	443	837	1095.83	60	18.2639	7.8219
8	11	328	697	1097.41	46	23.8567	7.2404
9	11	411	752	1048.8	49	21.4041	7.7131
10	7	293	713	762.056	35	21.773	6.3876
11	10	298	692	1718.16	65	26.4332	9.4006
12	8	253	391	452.404	24	18.8502	7.539
13	2	172	346	925.737	27	34.2866	7.6891
14	9	232	526	1017.21	41	24.81	6.6042
15	9	232	462	479.078	24	19.9616	5.9188
16	7	356	799	1455.63	80	18.1953	6.9684
17	9	261	497	1033.44	44	23.4872	6.7262

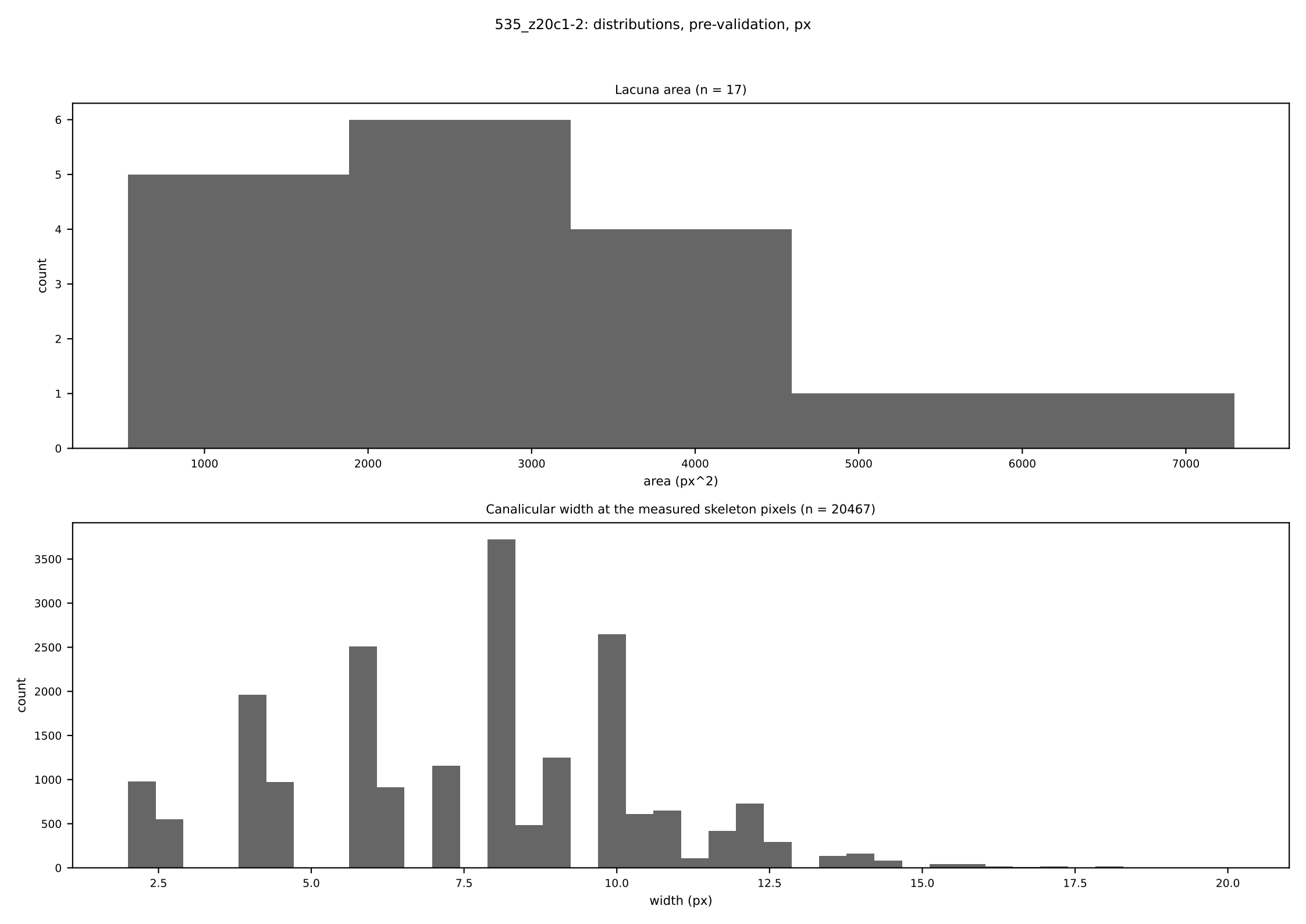
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.326	2172	4611	0.5445	0.5303	0.0234807	0.0455433	0
2	226.072	5807	13167	0.8341	0.734	0.035991	0.0268094	2.654
3	303.605	10283	25801	1	0.9954	0.0359817	0.0339134	2.9644
4	92.8112	5363	16377	1	1	0.0479209	0.0372474	5.3873
5	243.179	9488	24489	1	1	0.0478499	0.0403446	4.5234
6	208.894	7431	16405	1	1	0.0374109	0.031149	3.351
7	446.174	13512	24714	1	1	0.0327857	0.0338674	3.586
8	391.931	12275	26488	1	1	0.026721	0.0263138	2.8066
9	312.191	11238	24088	1	1	0.0365723	0.0312189	3.5235
10	198.995	8411	21092	1	1	0.0348353	0.0338043	3.5177
11	327.019	11681	28795	1	1	0.0255115	0.0240319	3.0579
12	223.965	8975	20144	1	1	0.0281894	0.0194103	3.572
13	139.681	5912	13025	1	1	0.0290934	0.0265643	1.4318
14	473.073	9278	19843	1	1	0.0250054	0.0265081	1.9025
15	421.339	9660	21532	1	1	0.0240166	0.0214564	2.136
16	262.149	9124	20333	1	1	0.039018	0.0392957	2.6702
17	177.782	8012	21351	1	1	0.0325761	0.0232776	5.0624

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	11	11	57.7696	240.38	1	0	4
2	189	253	238.995	403.948	9	5	4
3	350	742	422.505	1001.06	8	13	9
4	247	542	303.563	706.926	5	10	9
5	446	949	516.718	1113.68	12	10	11
6	278	472	322.321	590.357	6	9	10
7	364	601	502.375	950.021	8	14	16
8	260	520	366.635	788.139	12	9	8
9	362	692	469.647	869.737	8	13	12
10	182	304	337.392	807.269	7	8	13
11	220	465	343.948	781.583	6	11	10
12	250	380	283.451	444.647	7	9	6
13	114	222	196.924	387.735	2	7	5
14	230	306	265.066	602.973	7	3	5
15	221	305	253.439	515.404	6	3	4

16	356	605	409.12	916.394	6	10	9
17	256	415	289.581	560.032	7	6	6


535_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 20467)
count
width (px)